

Performance Testing

Date	2 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance testing checks how fast and how stable EduGenie stays when several learners use it at the same time. The load test was run locally with Locust, and the laptop's CPU and memory were recorded in Task Manager.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python load-testing tool) with a locustfile.py script; Windows Task Manager for CPU and memory
Type of Testing	Load Testing (moderate concurrent use)
Target Module / API	/qa, /explain, /quiz, /summarize and /learn/recommendations endpoints, and the home page
Test Environment	Local laptop: Intel Core i5-13420H, 16 GB RAM, Windows, Python 3.11. EduGenie ran on Uvicorn at http://127.0.0.1:8000 and called the Google Gemini API over the internet.
Test Date	2 October 2026 (final run about 7:55 to 7:58 AM)

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Mixed learner traffic: Locust users randomly open the home page and use Q&A, concept explanation, quiz generation, summarization and learning-path requests together.	10	about 150	All endpoints respond without errors, with AI responses in under 10 seconds on average.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Avg Response Time, home page	< 2 seconds	9.66 ms (0.01 s)	PASS	Far below the target.
2	Avg Response Time, AI endpoints	< 10 seconds	about 805 ms (0.8 s)	PASS	Average of the 5 AI endpoints.
3	Max Response Time, AI endpoints	< 20 seconds	12.634 s	PASS	One slow /learn/recommendations call.
4	Throughput (requests / sec)	Stable, no drop	4.3 req/s	PASS	Held near 4.3; one brief dip recovered.
5	Error Rate	< 5%	15.93% (72 of 452)	FAIL	All errors were HTTP 500 from /quiz.
6	CPU Utilization	< 80%	8%	PASS	Peaks stayed under about 20%.
7	Memory Utilization	< 80%	55% (8.6 of 15.6 GB)	PASS	Whole system, not only EduGenie.

Results by endpoint (from the Locust Statistics tab)

Endpoint	Requests	Failures	Median (ms)	95th % (ms)	Average (ms)	Max (ms)
/explain	43	0	360	3300	893	5440
/learn/recommendations	82	0	350	5500	1068	12634
/qa	145	0	360	2700	728	10467
/quiz	80	72	350	2800	607	6675
/summarize	67	0	360	4800	828	5731
Home page (/)	35	0	7	29	10	31
All requests	452	72	350	3700	743	12634

Step 4: Observations & Analysis

- **Load handled:** with 10 virtual users, Locust sent 452 requests in about 2.5 minutes at roughly 4.3 requests per second.
- **Speed:** the home page averaged 9.66 ms and the AI endpoints about 805 ms, both well inside the targets. /learn/recommendations was the slowest endpoint (average 1.07 s, one request of 12.6 s) because it generates the longest answer.
- **Errors:** /quiz returned HTTP 500 for 72 of its 80 requests. Every other endpoint had zero failures, so the overall error rate of 15.93% missed the 5% target. The Failures tab was captured moments later and shows 86 failures as the run continued. The exact cause is being checked in the server log; likely candidates are quiz output from the AI that fails validation, or API rate limiting under concurrent requests.
- **Resources:** CPU stayed at about 8% and memory at 55% for the whole laptop, so the server is not resource-bound. Most of the time is spent waiting for the Gemini API.
- **Result:** 6 of 7 metrics passed. Next step: fix the /quiz errors (for example by retrying when the quiz output is invalid) and repeat the test.

Step 5: Screenshots / Evidence

All screenshots are from the final 10-user mixed-load test on 2 October 2026.

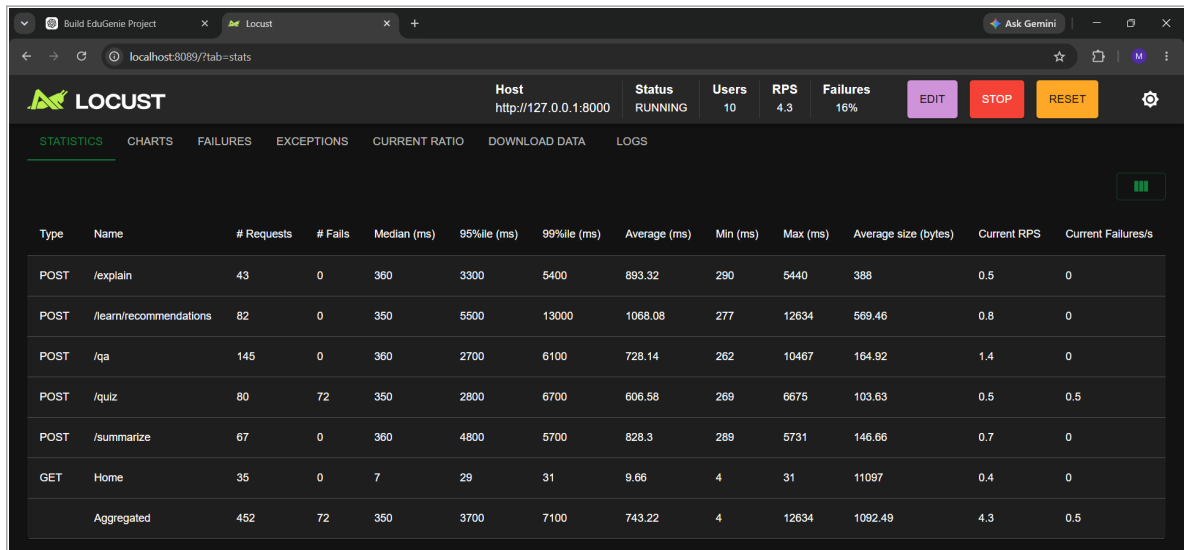


Fig. 1 Locust Statistics: requests, failures and response times per endpoint

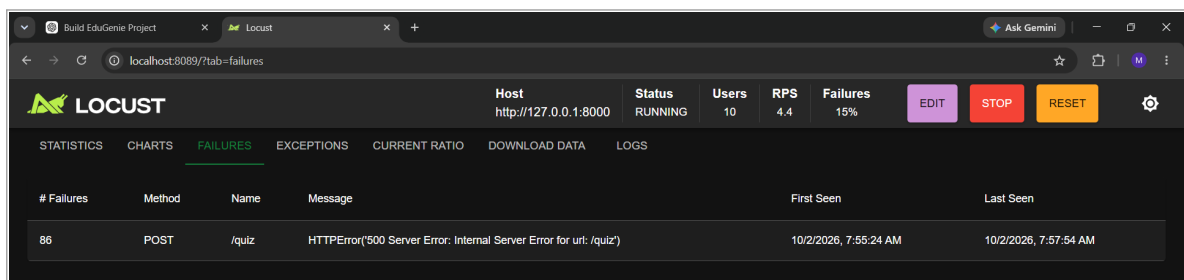


Fig. 2 Locust Failures: HTTP 500 errors from POST /quiz (captured moments after Fig. 1, so the count is higher)

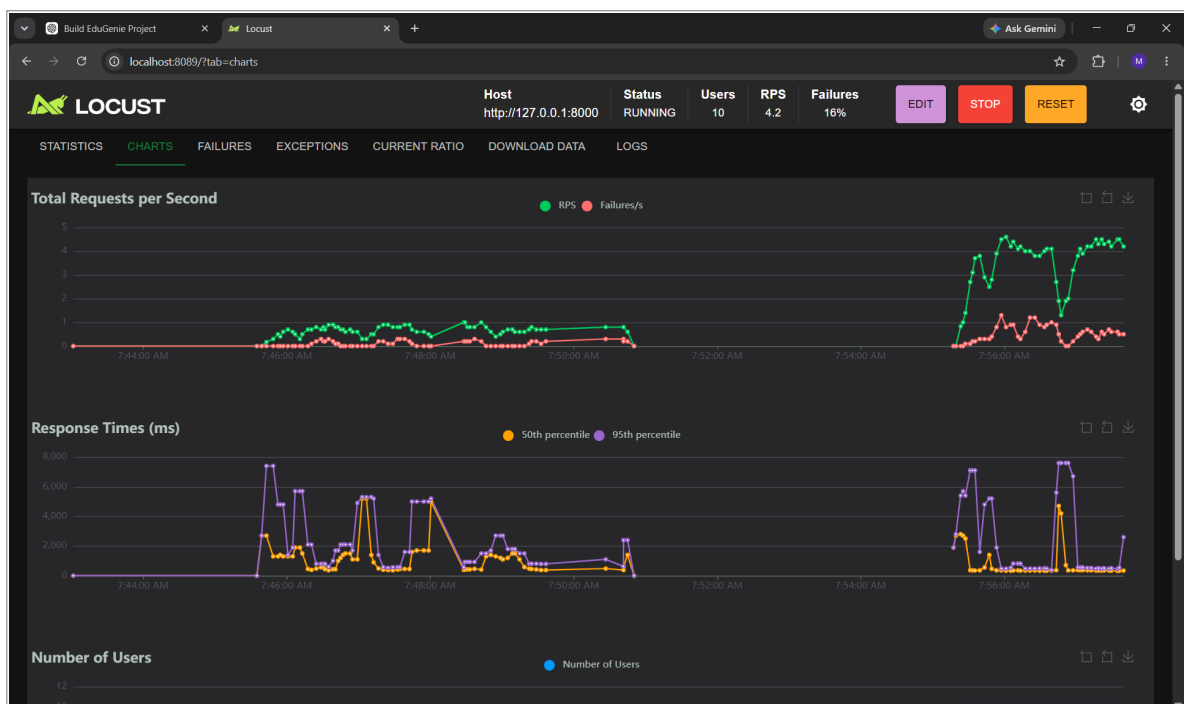


Fig. 3 Locust Charts: requests per second, response times and number of users

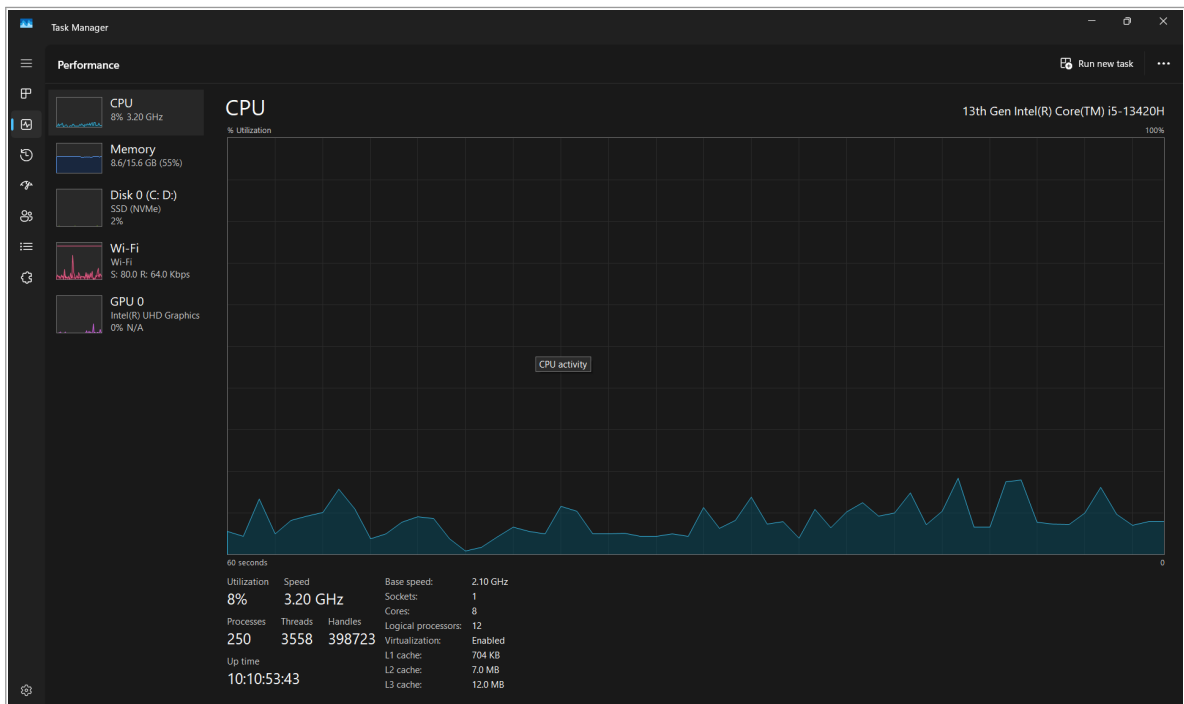


Fig. 4 Task Manager: CPU utilization during the test (8%)

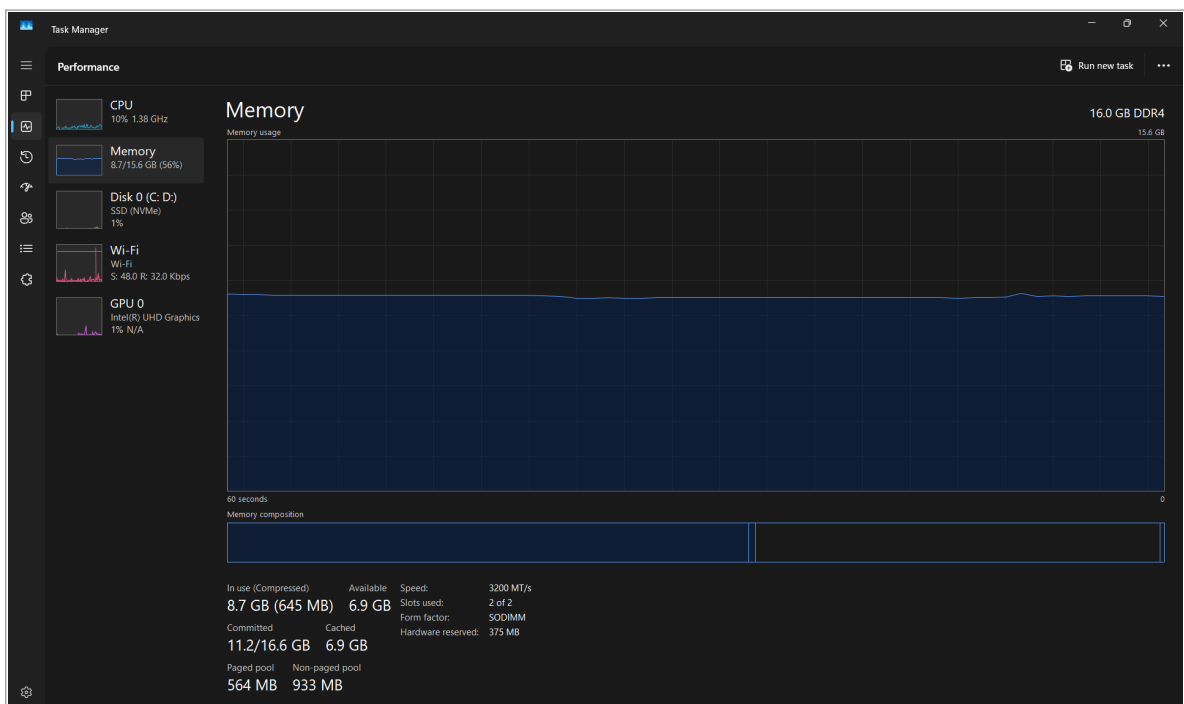


Fig. 5 Task Manager: memory utilization during the test (8.7 of 15.6 GB, 56%)